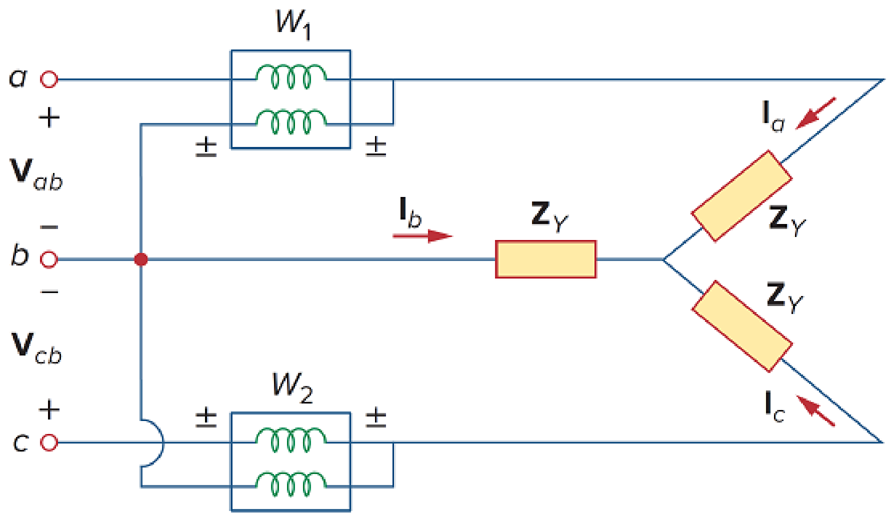
Chapter 12, Problem 15PP

Problem

If the load in Fig. 12.35 is delta-connected with impedance per phase of **Z***p* = 30 + *j*40 Ω and *VL* = 220 V, predict the readings of the wattmeters *W*1 and *W*2. Calculate *PT* and *QT*.

Figure 12.35



Step-by-step solution

Step 1 of 6

Convert the delta connected load to wye connection.

Calculate the impedance of the wye connected load.

$$Z_Y = \frac{Z_\Delta}{3}$$

Substitute $30 + j40\ \Omega$ for Z_Δ.

$$\begin{aligned} Z_Y &= \frac{30 + j40}{3} \\ &= \frac{50\angle 53.13^\circ}{3} \\ &= 16.667\angle 53.13^\circ\ \Omega \end{aligned}$$

Therefore, the impedance of the wye connected load is, $16.667\angle 53.13^\circ\ \Omega$.

The magnitude of the load $|Z_Y|$ is $16.667\ \Omega$ and the phase angle θ° is 53.13°.

Step 2 of 6

Refer to Figure 12.35 in the text book.

Calculate the magnitude value of the line current I_L.

$$I_L = \frac{V_L}{\sqrt{3}|Z_Y|}$$

Substitute $16.667\angle 53.13°\ \Omega$ for $|Z_Y|$ and $220\ V$ for V_L.

$$\begin{aligned} I_L &= \frac{220}{\sqrt{3}(16.667)} \\ &= \frac{220}{28.868} \\ &= 7.621\ A \end{aligned}$$

Therefore, the magnitude value of the line current I_L is $7.621\ A$.

Step 3 of 6

Calculate the value of the power reading in wattmeter *W*1.

$$W_1 = V_L I_L \cos(\theta° + 30°)$$

Substitute $220\ V$ for V_L, $7.621\ A$ for I_L and $53.13°$ for $\theta°$.

$$\begin{aligned} W_1 &= (220)(7.621)\cos(53.13° + 30°) \\ &= (1676.62)\cos(83.13°) \\ &= (1676.62)(0.1196) \\ &= 200.52\ W \end{aligned}$$

Therefore, the value of the power reading in wattmeter *W*1 is $\boxed{200.52\ W}$.

Step 4 of 6

Calculate the value of the power reading in wattmeter *W*2.

$$W_2 = V_L I_L \cos(\theta° - 30°)$$

Substitute $220\ V$ for V_L, $7.621\ A$ for I_L and $53.13°$ for $\theta°$.

$$\begin{aligned} W_2 &= (220)(7.621)\cos(53.13° - 30°) \\ &= (1676.62)\cos(23.13°) \\ &= (1676.62)(0.9196) \\ &= 1.5418\ kW \end{aligned}$$

Therefore, the value of the power reading in wattmeter *W*2 is $\boxed{1.5418\ kW}$.

Step 5 of 6

Calculate the value of the total real power P_T absorbed by the load.

$$P_T = W_1 + W_2$$

Substitute $1.5418\ kW$ for W_2 and $200.52\ W$ for W_1.

$$\begin{aligned} P_T &= 200.52 + 1.5418\ k \\ &= 1.7423\ kW \end{aligned}$$

Therefore, the value of the total real power P_T absorbed by the load is $\boxed{1.7423\ kW}$.

Step 6 of 6

Calculate the value of the total reactive power Q_T absorbed by the load.

$$Q_T = \sqrt{3}\left(W_2 - W_1\right)$$

Substitute $1.5418\ \mathrm{kW}$ for W_2 and $200.52\ \mathrm{W}$ for W_1.

$$\begin{aligned} Q_T &= \sqrt{3}\left(1.5418\ \mathrm{k} - 200.52\right) \\ &= \sqrt{3}\left(1.3413\ \mathrm{k}\right) \\ &= 2.3232\ \mathrm{kVAR} \end{aligned}$$

Therefore, the value of the total reactive power Q_T absorbed by the load is $\boxed{2.3232\ \mathrm{kVAR}}$.